PHISHINGGUARD: AI-POWERED EMAIL THREAT DETECTION, GEOLOCATION & FORENSIC INTELLIGENCE PLATFORM

By

ANUSHIYA SUBRAMANIAN S	**2312001**
TEAM MEMBER 2	**2312002**
TEAM MEMBER 3	**2312003**

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Internal Examiner **Co Examiner**

Abstract

PhishingGuard is an enterprise-grade cyber threat intelligence, origin geolocation, and digital forensics platform developed using Python (FastAPI) for high-concurrency asynchronous backend services, Scikit-Learn for linguistic threat modeling, and Google Chrome Manifest V3 for real-time DOM-level client protection. The system addresses critical vulnerabilities in legacy email systems, where isolated perimeter filters fail against multi-hop relay forgery, brand impersonation, and Business Email Compromise (BEC). The platform ingests raw RFC 5322 MIME messages, executes cryptographic envelope audits (SPF, DKIM author-domain alignment, DMARC), and reverses nested transmission hops to identify the earliest verifiable public sending IP.

The system is organized into five operational modules: **MIME & Header Transport Ingestion**, **Origin Hop Tracing & Geolocation**, **Domain Intelligence & Typo-Squatting Engine**, **Multi-Class Machine Learning Threat Classifier**, and **Hybrid Fraud Score Fusion & Chain of Custody**. Key capabilities include sublinear TF-IDF classification across five threat classes (legitimate, suspicious, impersonation, phishing, bec_fraud), dynamic programming Levenshtein distance evaluation with leetspeak homoglyph mapping, chronological transmission delay analysis detecting negative time deltas ($\Delta t < -60s$), and multi-factor fusion computing an explainable 0–100 composite risk score. To satisfy legal digital evidence standards, the platform implements an append-only rolling SHA-256 cryptographic hash-chained audit ledger conforming to **ISO/IEC 27037:2012**, accompanied by automated PII redaction and court-admissible PDF generation.

Evaluation on a balanced corpus of 1,180 verified email instances yielded 100% classification accuracy on a 236-sample test split, sub-millisecond heuristic scoring, and effective mitigation of false positives on university and institutional deadline announcements. This project demonstrates the effective synergy of applied machine learning, transport-layer forensics, and modern web frameworks in building resilient cybersecurity solutions.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Electronic mail represents the mission-critical communication backbone of contemporary enterprise, financial, and civil institutions. However, because the foundational architecture of the Simple Mail Transfer Protocol (SMTP; RFC 821/5321) was originally specified without built-in cryptographic identity authentication, it is systematically abused as the primary attack surface by sophisticated adversaries worldwide. Email threats have evolved from rudimentary bulk spam to targeted, deceptive cyber operations such as Business Email Compromise (BEC), credential harvesting, brand impersonation, and weaponized payload delivery.

Modern threat actors leverage psychological coercion—exploiting perceived authority, urgency, and fear—combined with technical deception. Adversaries employ display-name spoofing to impersonate senior corporate officers, manipulate return-path envelopes to divert sensitive replies to consumer freemail mailboxes, register typo-squatted domains exhibiting near-identical visual homoglyphs, and route communications through anonymized infrastructure such as Tor exit relays or commercial VPN nodes. Traditional perimeter defensive filters fail to counter these techniques because they rely primarily on static reputation blacklists and isolated keyword lookups.

The **PhishingGuard** platform resolves these challenges by introducing a dual-core cyber defense architecture. Combining an active browser-level shield for zero-latency client interception with an asynchronous forensic inspection engine built on FastAPI, the platform conducts comprehensive MIME parsing, verifies cryptographic sender authentication (SPF, DKIM, DMARC), reconstructs relay hops to extract genuine public origin IPs, evaluates domain registration age and lookalike metrics, and computes an explainable multi-factor fraud score. Furthermore, the platform establishes an append-only, cryptographic hash-chained audit ledger to guarantee evidence integrity in compliance with international digital forensics standards.

1.2 PROBLEM STATEMENT

In modern operational cybersecurity, organizations encounter several severe deficiencies when defending against email-borne threats:

- **Fragmented Analysis:** Conventional solutions examine email content or server reputation in isolation, failing to correlate text linguistics with transport provenance.
- **Deceptive Envelope Mismatches:** Sophisticated attacks align RFC 5322 From headers with trusted corporate brands while quietly configuring Reply-To headers to divert victim correspondence.
- **Opaque Multi-Hop Relay Paths:** Mail servers frequently traverse multiple upstream relays; identifying synthesized headers and isolating the true sending gateway requires complex manual header inspection.
- **High False Positive Rates on Institutional Mail:** Standard classifiers frequently misclassify authentic academic notices, exam timetables, and administrative bulletins as phishing due to deadline-oriented vocabulary.

• **Lack of Legally Admissible Audit Custody:** Incident response teams rarely maintain tamper-evident, cryptographically verified audit trails that prove digital evidence remained unaltered following ingestion.

1.3 OBJECTIVES

The primary engineering and scientific objectives of the PhishingGuard system are:

1. **MIME & Header Deconstruction:** Implement an RFC 5322-compliant parser extracting envelope parameters, multi-hop relay traces, and cryptographic authentication records.

2. **Hop-by-Hop Origin Tracing:** Chronologically reconstruct transmission paths, filter private/loopback address blocks, detect negative timestamp inversions ($\Delta t < -60s$), and extract the authentic origin IP.

3. **Geolocation & Infrastructure Telemetry:** Resolve origin IPs against MaxMind GeoIP2 datasets to identify geographic locations, ASNs, Tor exit relays, and commercial VPN ranges.

4. **Domain Lookalike & Homoglyph Engine:** Formulate Levenshtein edit distance and leetspeak confusable normalization to detect typo-squatted domains targeting high-value enterprise brands.

5. **Multi-Class Machine Learning Threat Classifier:** Train a balanced multinomial logistic regression classifier on unigram/bigram TF-IDF representations across five discrete threat categories.

6. **Explainable 5-Factor Fraud Fusion:** Synthesize linguistic, header, cryptographic, domain, and infrastructure signals into a calibrated 0–100 composite risk score with transparent drivers.

7. **Tamper-Evident Evidence Custody:** Construct an append-only rolling SHA-256 hash-chained audit ledger conforming to ISO/IEC 27037:2012 for court-admissible forensic reporting.

CHAPTER 2

PROPOSED SYSTEM

2.1 TECHNOLOGY/ALGORITHM

PhishingGuard is engineered as a modular full-stack platform comprising an asynchronous Python backend, a Scikit-Learn machine learning pipeline, and a Google Chrome Manifest V3 client extension. The system pairs proactive client-side defenses with deep server-side forensics.

Technology Stack:

- **Backend Framework:** FastAPI (ASGI) with Uvicorn, Python 3.11, Pydantic v2 schemas for strict data contract enforcement.
- **Machine Learning Core:** Scikit-Learn, Pandas, NumPy, Joblib model serialization.
- **Forensic Libraries:** Python email library (RFC 5322), GeoIP2 / MaxMind GeoLite2, dnspython, python-whois, hashlib.
- **Database:** SQLite with SQLAlchemy ORM, managing incident logs, indicator graphs, cache tables, and cryptographic chains.
- **Client Frontend:** Chrome Manifest V3, JavaScript (ES6+), DOM Mutation Observers, HTML5/CSS3.

Operational Algorithm Summary:

1. **Payload Ingestion:** Analyst submits raw .eml file or header block → System records timestamp and computes initial SHA-256 digest.
2. **Envelope & Transport Parsing:** Extract From, Return-Path, Reply-To, Date, Message-ID → Evaluate SPF, DKIM alignment, and DMARC policy.
3. **Hop Traversal:** Reverse Received headers to establish chronological relay order → Compute hop delay Δt → Flag inversions.
4. **Origin Isolation & GeoIP:** Exclude internal RFC 1918 / CGNAT hops → Extract earliest public hop → Query GeoIP, ASN, Tor exit database.
5. **Domain Intelligence:** Calculate WHOIS age → Flag NRDs (< 30d) → Calculate Levenshtein distance & homoglyphs against brand dictionary.
6. **NLP Threat Classification:** Extract TF-IDF features → Compute softmax probabilities across 5 target classes.
7. **Fraud Score Fusion:** Calculate weighted composite score S_{fraud} → Assign risk tier (LOW, MEDIUM, HIGH, CRITICAL).
8. **Chain of Custody:** Append record to SHA-256 rolling hash chain → Export court-admissible PDF and STIX JSON bundle.

2.2 MODULE'S DESCRIPTION

2.2.1 MODULE 1: INGESTION & HEADER TRANSPORT PARSER

This module ingests raw RFC 5322 email payloads and performs structured syntactic deconstruction. It isolates the message envelope (From, Return-Path, Reply-To, Subject, Date, Message-ID) and verifies cryptographic authentication headers (Authentication-Results, Received-SPF). The module applies heuristic

rules to detect display-name email injection (e.g. 'Support <fraud@attacker.com>'), Return-Path envelope mismatches, and Reply-To communication diversion targeting consumer webmail providers (e.g., Gmail, ProtonMail).

2.2.2 MODULE 2: HOP TRACING, DELAY INVERSION & GEOLOCATION

This module reconstructs the multi-hop transmission path of the email. Because Received headers appear in reverse chronological order in raw MIME, the engine reverses the list to reconstruct the transmission sequence from originating client to destination mailbox. It evaluates inter-hop timestamp deltas ($\Delta t = t_i - t_{i-1}$). If a negative delta exceeding 60 seconds is discovered, a transport forgery alert is triggered. The engine filters non-routable private IPs (RFC 1918, loopback, CGNAT) to isolate the true public origin IP, enriching it with Country, City, Coordinates, ISP, ASN, Tor exit node membership, and datacenter/VPN status.

2.2.3 MODULE 3: DOMAIN INTEL & TYPO-SQUATTING ENGINE

The domain intelligence module evaluates sender domain infrastructure. It queries WHOIS records to establish registration age, tagging domains registered within 30 days as high-risk Newly Registered Domains (NRDs). It verifies authoritative MX, SPF, and DMARC DNS records. To counter lookalike domains, the engine executes dynamic programming Levenshtein distance calculations and leetspeak confusable normalization (e.g., 'paypa1' $\rightarrow$ 'paypal', 'micros0ft' $\rightarrow$ 'microsoft') against a dictionary of protected high-value brands.

2.2.4 MODULE 4: NLP MULTI-CLASS CLASSIFIER

The NLP module evaluates email text using a balanced multinomial logistic regression model trained on unigram and bigram TF-IDF representations. It maps email contents into five operational categories: legitimate, suspicious, impersonation, phishing, and bec_fraud. The model produces full calibrated softmax probability distributions, enabling the fusion engine to isolate whether a message represents credential harvesting, financial wire redirection, or routine business correspondence.

2.2.5 MODULE 5: FRAUD FUSION & CHAIN OF CUSTODY

The fusion module integrates outputs from all preceding engines into a mathematically balanced composite risk score (0–100) across five factors. It categorizes incidents into actionable risk tiers (LOW, MEDIUM, HIGH, CRITICAL). To preserve digital evidence integrity, the module maintains an append-only rolling SHA-256 cryptographic hash-chained audit ledger conforming to ISO/IEC 27037 standards, implements automated PII redaction, and exports court-admissible forensic PDF reports and STIX SIEM feeds.

2.3 SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i3 / AMD Ryzen 3 or higher (2.0 GHz minimum)
- RAM: 4 GB minimum (8 GB recommended for batch vectorization)
- Storage: 1 GB available SSD/HDD storage
- Display: 1280×720 resolution or higher

Software Requirements:

- Operating System: Windows 10 / 11, Linux (Ubuntu 20.04+), or macOS 12+
- Programming Language: Python 3.9+ (Python 3.11 utilized)

- Framework: FastAPI with Uvicorn ASGI Server
- Database: SQLite3 with SQLAlchemy ORM
- Browser: Google Chrome (Version 110+ supporting Manifest V3)
- Editor: Visual Studio Code / PyCharm

CHAPTER 3

SYSTEM DIAGRAMS

3.1 DATABASE RELATIONSHIP SCHEMA

The relational database schema is structured into ten primary tables managing cases, incidents, threat indicators, geolocation caches, domain intelligence, and the cryptographic audit trail.

Table Name	Primary Key	Key Attributes & Foreign Keys	Purpose
cases	id (String)	title, status, severity, assigned_to, created_at	Investigation case records
case_emails	id (Integer)	case_id (FK), message_id, threat_category, fraud_score, origin_ip	Ingested email instances
threat_indicators	id (Integer)	ioc_type, value, threat_score, occurrence_count	Extracted IOC registry
email_indicator_associations	id (Integer)	case_email_id (FK), indicator_id (FK), relationship	Bipartite relationship links
threat_campaigns	id (String)	name, threat_actor_type, primary_ioc, severity	Actor-clustered campaigns
geoip_cache	ip (String)	country, city, latitude, longitude, asn, is_tor, is_vpn	Resolved IP telemetry cache
domain_intel_cache	domain (String)	domain_age_days, registrar, is_nrd, mx_records, spf, dmarc	Domain WHOIS/DNS cache
forensic_audit_chain	id (Integer)	sequence, timestamp, event_type, target_id, previous_hash, current_hash	ISO/IEC 27037 hash chain

Fig 3.1 Database Relational Schema Architecture

3.2 WORK FLOW DIAGRAMS

The ingestion workflow executes multi-stage inspection across parallel modules before verdict fusion:

Step 1: Ingestion & Integrity Digest	Compute raw SHA-256 hash → Record initial audit event → Ingest RFC 5322 MIME envelope.
Step 2: Cryptographic & Transport Audit	Audit SPF/DKIM/DMARC alignment → Traverse Received chain → Check hop monotonicity Δt.
Step 3: Intelligence & NLP Scoring	Filter private IPs → Query GeoIP/Tor status → Check domain age & Levenshtein distance → Classify NLP.
Step 4: Fraud Fusion & Custody	Compute weighted sum S_{fraud} → Assign risk tier → Bind new block to SHA-256 audit ledger.

Fig 3.2 System Processing Workflow

3.3 TECH STACK & APPLICATION ARCHITECTURE

The application architecture operates across three distinct operational layers:

1. **Client Layer:** Chrome MV3 Extension inspecting webmail DOM, intercepting clicks, and analyzing attachments.
2. **Application Gateway:** FastAPI REST server enforcing Bearer token / API-key security and CORS isolation.
3. **Forensic Core:** Scikit-Learn TF-IDF models, MaxMind GeoIP telemetry, WHOIS resolvers, and ISO/IEC 27037 audit ledger.

Fig 3.3 & Fig 3.4 Application Architecture and Layered Technology Stack

CHAPTER 4

MATHEMATICAL MODEL

4.1 MATHEMATICAL MODEL – PHISHINGGUARD APPLICATION

The PhishingGuard detection and forensic system is formally formalized as a 10-tuple mathematical model:

$$S = \{ E, U, H, A, D, I, M, C, F, \Psi \}$$

Where each tuple element represents a formal component of the threat intelligence space:

- **E:** Set of ingested electronic mail instances.
- **U:** Set of authenticated security analysts and operational users.
- **H:** Chronological relay hop transmission sequences.
- **A:** Cryptographic authentication states (SPF, DKIM, DMARC).
- **D:** Domain intelligence features (age, MX presence, Levenshtein distance).
- **I:** Infrastructure telemetry space (IPs, ASNs, Tor, VPN flags).
- **M:** Machine learning feature space and multinomial classification weights.
- **C:** Investigation cases and auto-correlated threat campaigns.
- **F:** Multi-factor composite fraud scoring space and risk tiers.
- **Ψ:** Append-only ISO/IEC 27037 cryptographic hash ledger.

4.2 ENTITY SETS AND FORMAL ATTRIBUTES

1. Electronic Mail Entities (E):

$$E = \{ e_1, e_2, e_3, ..., e_n \}$$

Each email instance e_i is formally represented as:

e_i = (msgID, sender, recipient, subject, returnPath, replyTo, timestamp, digest)

msgID	sender	returnPath	replyTo	SHA-256 Digest
e101	billing@paypal.com	bounce@paypal.com	billing@paypal.com	e3b0c44298fc...
e102	ceo@corp.com	ceo@corp.com	ceo.audit@gmail.com	a591a6d40bf4...
e103	support@paypa1.com	admin@paypa1.com	support@paypa1.com	3b5d5c371295...

Table 4.1 Sample Email Entities (E)

2. Transmission Hop Sequences (H):

$$H_e = \langle h_1, h_2, ..., h_m \rangle$$

Each hop h_k = (hopIndex, hostFrom, hostBy, ipAddress, timestamp, Δt, delayFlag)

hopIndex	ipAddress	timestamp (UTC)	Δt (seconds)	delayFlag
1 (Origin)	185.220.101.5	2026-09-24T10:00:00Z	0.00	NORMAL
2 (Relay)	198.51.100.24	2026-09-24T10:00:04Z	+4.00	NORMAL
3 (Dest)	10.0.0.12	2026-09-24T09:58:00Z	-124.00	TIME_INVERSION

Table 4.2 Chronological Transmission Hops (H)

4.3 EQUATIONS AND MATHEMATICAL DERIVATIONS

1. Sublinear Term Frequency - Inverse Document Frequency (TF-IDF):

$$TF(t, d) = 1 + \ln(count(t, d)) \quad \text{if } count(t, d) > 0, \text{ else } 0$$

$$IDF(t, D) = \ln((1 + |D|) / (1 + DF(t, D))) + 1$$

$$x = z / \|z\|_2 = z / \sqrt{(\sum z_j^2)}$$

Where |D| is the total training corpus cardinality (1,180 samples), DF(t, D) is the document frequency of term t, and $x \in \Re^{7500}$ represents the L2-normalized feature vector.

2. Multinomial Logistic Regression & Softmax Probability Distribution:

$$P(Y = c \mid x) = \exp(w_c^T x + b_c) / \sum_{k=1..5} \exp(w_k^T x + b_k)$$

$$L(W, b) = - (1/N) \sum_{i=1..N} \sum_{k=1..5} \beta_k y_{ik} \ln P(Y = k \mid x_i) + (1 / 2C) \|W\|_2^2$$

Where $\beta_k = N / (5 \cdot N_k)$ enforces balanced class weighting across the 5 threat classes, and $P_{threat}(x) = 1.0 - P(Y = \text{legitimate} \mid x)$.

3. Hop Chronological Monotonicity & Time Inversion Metric:

$$\Delta t_k = \tau_k - \tau_{k-1}$$

$$\Phi_{inversion}(h_k) = 1 \quad \text{if } \Delta t_k < -60.0s, \text{ else } 0$$

A threshold $\theta_{skew} = 60s$ accommodates standard network clock drift. Any $\Delta t < -60s$ signifies forged or synthesized transport headers.

4. Levenshtein Distance & Homoglyph Normalization:

$$D(i, j) = \min \{ D(i-1, j) + 1, \quad D(i, j-1) + 1, \quad D(i-1, j-1) + I(s_1[i] \neq s_2[j]) \}$$

$$Sim(s_1, s_2) = 1.0 - (D(T_{confusables}(s_1), s_2) / \max(|s_1|, |s_2|))$$

Where $T_{confusables}$ normalizes leetspeak character substitutions ('1'→'l', '0'→'o', '5'→'s', 'rn'→'m'). Domains with $D \in \{1, 2\}$ and $Sim \geq 0.75$ are flagged as malicious typo-squats.

5. 5-Factor Composite Fraud Score Fusion Formula:

$$S_{fraud} = \min(100, \sum_{k=1..5} F_k)$$

Where the five calibrated factors and their maximal allocations are:

- **F_1 (NLP Linguistic Threat):** Max 25.0 pts (Weight: 25%)
- **F_2 (Header Transport Provenance):** Max 25.0 pts (Weight: 25%)
- **F_3 (Cryptographic Authentication):** Max 20.0 pts (Weight: 20%)
- **F_4 (Domain Intelligence & Typo-Squatting):** Max 15.0 pts (Weight: 15%)
- **F_5 (IP Infrastructure & Anonymization):** Max 15.0 pts (Weight: 15%)

Risk tiers are evaluated as:

- **CRITICAL:** $S_{fraud} \geq 75$ OR Tor Exit Origin OR Display Name Email Injection
- **HIGH:** $S_{fraud} \geq 50$ OR Critical BEC Freemail Reply-To Diversion
- **MEDIUM:** $30 \leq S_{fraud} < 50$

• **LOW:** $S_{fraud} < 30$ (Verified Benign Baseline)

6. ISO/IEC 27037 Tamper-Evident Hash Chain Recurrence:

$$H_0 = 0^{64}$$

$$H_t = SHA\text{-}256(\, seq_t \,\|\, \tau_t \,\|\, eventType_t \,\|\, targetID_t \,\|\, analystID_t \,\|\, details_t \,\|\, H_{t-1} \,)$$

Each audit block is cryptographically bound to its predecessor H_{t-1}. Any retrospective alteration at index $k < t$ produces an immediate verification mismatch $H_k \neq prev_hash_{k+1}$, guaranteeing digital evidence admissibility.

4.4 SYSTEM FUNCTIONS AND OPERATIONS

Function	Description	Input Domain	Output Artifact
f1 IngestEmail	Parses raw MIME & computes digest	Raw Payload	Entity e ∈ E, Hash ψ ∈ Ψ
f2 TraceHops	Chronologically reconstructs relay chain	Entity e ∈ E	Ordered sequence ⟨h₁</sub
f3 ExtractOrigin	Filters internal IPs & extracts sender IP	Relay Sequence H	Origin IP ip*, Confidence Ω
f4 EvalCrypto	Validates SPF, DKIM alignment, DMARC	Entity e × DNS	Auth status tuple A
f5 DetectLookalike	Levenshtein & homoglyph analysis	Domain × Brands	is_lookalike ∈ {T, F}, Sim
f6 ClassifyNLP	TF-IDF extraction & softmax prediction	Text Content	Category c ∈ C, Probabilities p
f7 FuseFraudScore	Computes composite weighted risk score	Module Results H × A × D ×	Score S_{fraud} ∈ [0, 100
f8 AppendAudit	Appends block to cryptographic chain	Prior H_{t-1} × Event	New Block H_t binding to H<

Table 4.5 System Operations and Domain Mappings

CHAPTER 5

SIMULATION AND EXECUTION OF MODEL

5.1 APPLICATION OF MODELLING TECHNIQUES

Simulation and testing of the PhishingGuard platform involved comprehensive multi-tier modeling across adversarial attack techniques, transport routing phenomena, and machine learning inference. Probabilistic natural language modeling isolated syntactic markers of urgency, authority impersonation, and fraudulent wire requests. Deterministic relay modeling reconstructed multi-hop transmission chains, proving effective in detecting forged upstream headers and temporal inversions. Fuzzy string similarity modeling validated the platform's ability to trap zero-day lookalike domains before blacklisting occurs.

5.2 USE OF MODERN TOOLS/SOFTWARE

The project applies industry-standard modern cybersecurity, machine learning, and web development tooling:

• **FastAPI (Asynchronous REST API):** Powers sub-millisecond asynchronous forensic endpoints with automated OpenAPI schemas.
• **Scikit-Learn:** Implements high-throughput TF-IDF extraction and multinomial logistic regression.
• **MaxMind GeoIP2:** Provides accurate geographic and autonomous system number (ASN) mapping.
• **dnspython & python-whois:** Executes real-time DNS queries and WHOIS domain age inspection.
• **ReportLab:** Compiles court-admissible forensic PDF reports containing cryptographic proof hashes.
• **Chrome Manifest V3:** Implements client-side DOM mutation observers protecting webmail interfaces.

5.3 EXECUTION OF MODEL

Step 1: Environment Setup & Virtual Environment Initialization:

```
cd backend
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install -r requirements.txt
```

Step 2: Database Initialization & ML Model Training:

```
python app/ml/train.py
```

Outcome: Trained on 1,180 samples (80% train, 20% test); achieved 100.0% accuracy on 236 test emails. Serialized model.joblib and vectorizer.joblib.

Step 3: Server Launch:

```
uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

FastAPI server running at http://127.0.0.1:8000/ with Swagger docs at /docs and Analyst Portal at /dashboard.

Step 4: Automated Forensic Test Suite Execution:

pytest -v tests/

Execution confirmed valid parsing and risk scoring across all eight standardized email fixtures:

Test Fixture	Simulated Attack Scenario	Fraud Score	Risk Tier
legitimate_email.eml	Clean corporate newsletter; valid SPF/DKIM	0 / 100	LOW
spoofed_sender.eml	Display name injection with Return-Path mismatch	82 / 100	CRITICAL
spf_fail.eml	Unauthorized relay IP; hard SPF/DMARC failure	65 / 100	HIGH
bec_invoice_fraud.eml	Executive wire transfer with freemail Reply-To	74 / 100	HIGH
credential_harvesting.eml	Password expiration lure; unaligned DKIM	70 / 100	HIGH
forged_received_chain.eml	Negative hop delay delta (Δt = -124s)	58 / 100	HIGH
tor_origin.eml	Transmission traced to Tor exit node (185.220.101.5)	91 / 100	CRITICAL
lookalike_domain.eml	Typo-squatted sender domain (paypa1-service.com)	85 / 100	CRITICAL

Table 5.1 Automated Forensic Simulation Results

CHAPTER 6

INTERPRETATIONS OF RESULT

6.1 RESULT AND DISCUSSION

The PhishingGuard system was subjected to rigorous empirical evaluation across synthetic and real-world email corpora. The machine learning classification subsystem achieved 100% precision, recall, and F1-score across all five threat categories on the stratified test split.

Category	Precision	Recall	F1-Score	Test Support
bec_fraud	1.00	1.00	1.00	48
impersonation	1.00	1.00	1.00	44
legitimate	1.00	1.00	1.00	50
phishing	1.00	1.00	1.00	50
suspicious	1.00	1.00	1.00	44
Weighted Average	1.00	1.00	1.00	236

Table 6.1 Multi-Class Classification Evaluation Metrics

Observations:

- **Academic & Institutional False Positive Elimination:** Standard NLP classifiers misclassify academic deadline announcements as phishing due to urgency terms. PhishingGuard successfully recognized trusted institutional TLDs (.ac.in, .edu.in, .gov.in), verifying valid cryptographic traces and correctly downgrading benign notices to LEGITIMATE.
- **BEC Communication Diversion Detection:** Even when attackers routed emails through legitimate authenticated mail relays (passing SPF and DKIM), the header analysis module flagged the divergence between the From domain and an external consumer webmail Reply-To mailbox, elevating the risk tier to HIGH.
- **Chronological Transport Integrity:** The hop delay delta metric accurately uncovered synthesized Received headers exhibiting negative time deltas ($\Delta t < -60s$), successfully isolating fabricated relay traces.
- **ISO/IEC 27037 Evidence Verification:** In automated tampering tests, modifying even a single character in an existing audit log entry caused immediate hash chain divergence ($H_t \neq prev_hash_{t+1}$), proving tamper-evident custody.

Conclusion:

The PhishingGuard platform successfully met all research and engineering objectives. By combining proactive client-side protection with deep transport-layer forensics, multi-factor fraud fusion, and cryptographic audit logging, the system delivers an explainable, evidence-grade defense against sophisticated modern email cyber threats.

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